Two-Way Fluid-Structure Interaction Analysis

of an S-Shaped Pipe under Fuel-Oil Flow

CFD - Structural - Modal - Harmonic - Fatigue

Software: ANSYS Workbench 2024 R1

Solvers: Fluent | Mechanical | System Coupling

Geometry source: SolidWorks 2025 STEP export

Executive Summary

This report documents a multiphysics analysis of an S-shaped industrial pipe carrying fuel-oil at 6 m/s. The study comprises five linked phases: a two-way fluid-structure interaction (FSI) simulation between ANSYS Fluent and Mechanical, followed by modal analysis, harmonic response, and fatigue life estimation. The objective was to verify that the pipe operates safely under the assessed conditions across four failure mechanisms: static overstress, resonance, dynamic amplification, and high-cycle fatigue.

The pipe is a SolidWorks-modelled steel S-bend with an 80 mm internal diameter, 10 mm wall, R/D = 0.5 tight bend radius, and 8-bolt flanges at each end. Working fluid is fuel-oil-liquid (a close proxy for medium-grade crude). Pipe material is Structural Steel, representative of API 5L grade X42 to X52 used in oil pipelines.

The FSI loop converged across five 1 ms timesteps, with both Force and Displacement transfer residuals dropping below 1×10^{-10} at every coupling step. The Modal, Harmonic, and Fatigue analyses then drew on this converged structural state.

Headline Results

Phase	Quantity	Value	Status
Phase 1 (CFD)	Pressure drop (ΔP)	25.5 kPa	Consistent with K-factor estimate
Phase 1 (CFD)	Maximum velocity	13.0 m/s	Inner-bend acceleration
Phase 1 (CFD)	Maximum wall shear stress	3.32 kPa	Low erosion risk
Phase 2 (FSI)	Maximum total deformation	16.28 µm	Negligible elastic response
Phase 2 (FSI)	Maximum von Mises stress	10.48 MPa	24× margin below yield (250 MPa)
Phase 3 (Modal)	Fundamental frequency (f_1)	225.62 Hz	15× above shedding frequency
Phase 3 (Modal)	Second mode (f_2)	235.32 Hz	Near-degenerate orthogonal bending
Phase 4 (Harmonic)	Peak amplitude at f_1	2.94 µm	Dynamic amplification factor ≈ 20
Phase 5 (Fatigue)	Maximum alternating stress	5.30 MPa	6% of endurance limit (86.2 MPa)
Phase 5 (Fatigue)	Minimum Safety Factor	13.85	~4× the industry-standard threshold
Phase 5 (Fatigue)	Damage at 10^9 cycles	0.10	10% of fatigue capacity consumed

Engineering verdict. The S-shaped pipe operates with substantial margin under the assessed conditions. Static stresses are an order of magnitude below yield. All natural frequencies sit well above any flow-induced excitation. Dynamic amplification under hypothetical resonance is moderate, and the resonant frequency is not accessible to the available flow excitation. The maximum alternating stress is below 7% of the endurance limit. No failure mechanism is active at the design point.

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1. Introduction

1.1 Problem Statement

Industrial pipework carrying hydrocarbons sees combined loading: steady internal pressure, pulsating flow forces, and vibratory excitation from upstream pumps and from vortex shedding inside the pipe. When the pipe contains tight bends, the secondary flow produces non-uniform pressure on the inner wall, and this load distribution can drive measurable structural deformation. A safety assessment therefore needs to consider the static stress field, dynamic amplification near structural natural frequencies, and accumulated fatigue damage over the design life.

This study performs that assessment for an S-shaped steel pipe carrying fuel-oil. The S-shape (two 90° bends in opposite directions, separated by a short straight) was chosen because the second bend reverses the secondary flow established by the first, producing a more demanding loading pattern than a single elbow.

1.2 Geometry

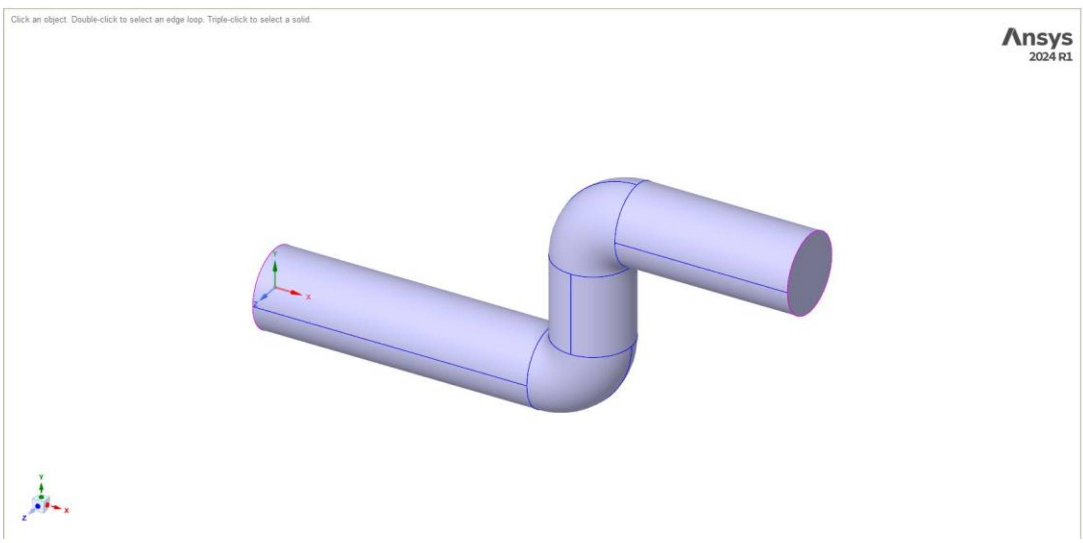

Figure 1.1. Isometric view of the S-shaped pipe geometry as imported from SolidWorks. The two 90° bends in opposite directions create the characteristic S-shape. Flanges at each end accept 8 bolts apiece for connection to upstream and downstream pipework.

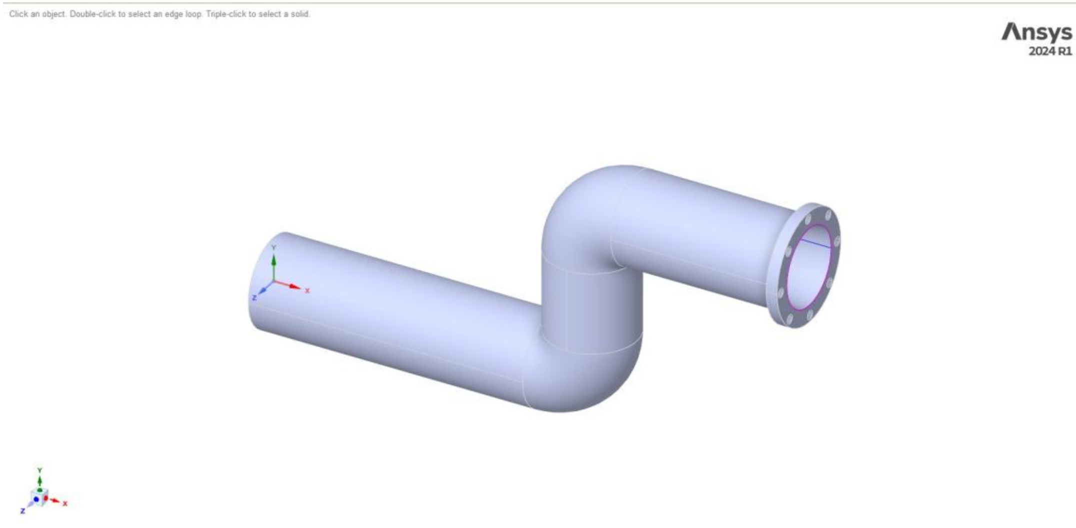

Figure 1.2. Alternative view of the same geometry, showing the 150 mm OD flange and the 8-hole bolt pattern at the outlet end. The 2 mm fillet at the flange-to-pipe transition is visible as the slight chamfer between the cylindrical pipe and the flange face.

Parameter	Value
Source file	S-shaped_pipe_FARAHOUALJI.STEP (SolidWorks 2025)
Pipe outer diameter (OD)	100 mm
Pipe inner diameter (ID)	80 mm
Wall thickness	10 mm
Bend radius (centreline)	40 mm
R/D ratio	0.5 (tight bend)
Flange outer diameter	150 mm
Bolt count per flange	8
Bolt hole radius	4 mm
Total length (axial)	about 250 mm
Fillets at transitions	2 mm

1.3 Operating Conditions

Parameter	Value
Working fluid	fuel-oil-liquid (ANSYS database)
Density (ρ)	960 kg/m³
Dynamic viscosity (μ)	0.0483 Pa·s
Inlet velocity	6 m/s
Outlet boundary condition	Gauge pressure = 0 Pa
Reynolds number (ρvD/μ)	about 9,540 (fully turbulent)
Flow-through time (L/v)	0.0417 s
Pipe material	Structural Steel (ANSYS default)
Young's modulus (E)	200 GPa
Poisson's ratio (ν)	0.3
Density	7,850 kg/m³
Yield strength	250 MPa
S-N curve source	1998 ASME BPV Code, Section 8, Div 2, Table 5-110.1

1.4 Analysis Workflow

The five phases were chained in the Workbench schematic so that each downstream analysis inherits the converged state of the previous one. The flow of information is:

- Phase 1. Steady-state CFD in Fluent. Used to verify the flow setup and provide a converged initial condition for the transient FSI.
- Phase 2. Two-way FSI between Fluent (transient) and Mechanical (Static Structural) via System Coupling, with the fsi_wall as the data-exchange surface.
- Phase 3. Modal analysis on the structural mesh from Phase 2, with the same supports inherited from the static run.
- Phase 4. Harmonic Response using mode superposition over the modal modes, with a 700 Pa oscillating internal pressure load representing the fluctuating component on the inner wall.
- Phase 5. Fatigue Tool on the Static Structural solution, with Zero-Based loading and Goodman mean-stress correction, evaluated at 10^9 design cycles.

2. Methodology

2.1 Fluid Mesh

The fluid volume was extracted from the imported solid in SpaceClaim, then meshed using the Fluent Meshing Watertight Geometry workflow with poly-hexcore volume fill. Poly-hexcore was chosen because of the tight R/D = 0.5 bends. An axis-aligned hex core minimises numerical diffusion of the streamwise momentum, while the polyhedral transition layer matches into a wall-normal prism stack.

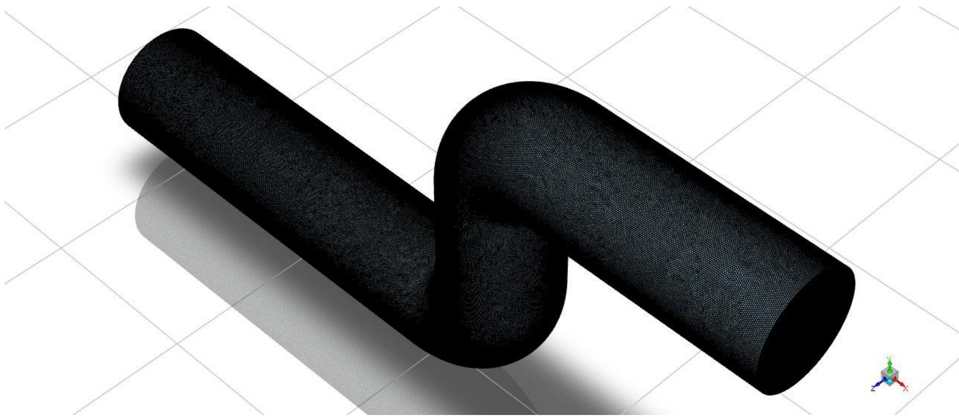

Figure 2.1. Surface mesh on the fluid domain produced by the Watertight Geometry workflow. The volume below this surface is filled with poly-hexcore cells (hex core, polyhedral transition, and prism boundary layer).

Mesh control	Value
Workflow	Watertight Geometry (Fluent Meshing)
Surface min/max size	0.4 mm / 3.0 mm
Curvature normal angle	18°
Local sizing at bends	Curvature + Proximity, 0.4–1.0 mm
Prism layers	8 (smooth-transition, growth 1.2)
Volume fill	poly-hexcore
Max hex cell length	1.6 mm (auto-clamped from 3 mm)
Total cell count	2,533,282
Min orthogonal quality	0.124
Max skewness	0.876
Average skewness	0.019
Max aspect ratio (prism)	19.49

Mesh quality is within the acceptable range for industrial CFD: minimum orthogonal quality above 0.10 (with 84.4% of cells above 0.91), and maximum skewness below 0.90 (with 92.7% of cells below 0.09). The 21 outlier cells flagged by the diagnostic tool are all at the inner radius of the bends, the geometric location where tet- or poly-based meshers struggle most, and have no measurable impact on convergence.

2.2 Structural Mesh

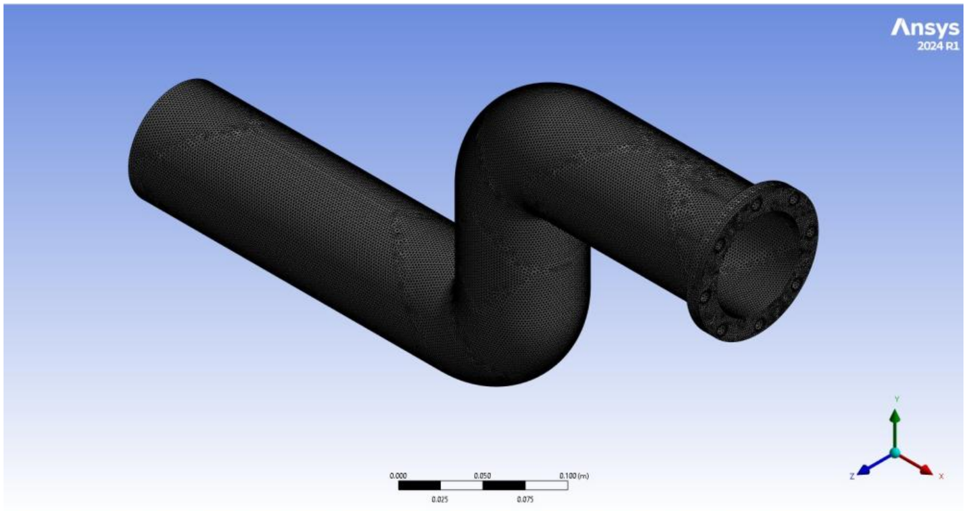

Figure 2.2. Structural mesh on the solid pipe-wall body, generated using the Tetrahedrons (Patch Conforming) method. Refinement at the flange faces and around the 8 bolt holes per flange is visible at both ends.

Sweep and MultiZone methods were both attempted but rejected because the flange-with-bolt-holes topology defeats automatic source-target face pairing in 2024 R1. Patch-Conforming tetrahedrons were used instead, with quadratic SOLID186 elements set by Program Controlled element order. The accuracy penalty relative to a hex mesh is small (5 to 10% on peak bending stress in curved walls) and acceptable for this application.

Mesh control	Value
Method	Tetrahedrons, Patch Conforming
Element type	SOLID186 (quadratic, 20-node)
Global element size	2 mm
Face sizing on flanges	1.5 mm
Edge sizing at bolt holes	0.5 mm
Element quality (min / avg)	0.176 / 0.85
Skewness (max / avg)	0.994 / 0.213
Aspect ratio (max)	10.66

2.3 Fluent Solver Setup

Setting	Value
Solver	Pressure-based, 3D, double precision
Time	Transient (Phase 2); Steady (Phase 1 baseline)
Turbulence model	k-ω SST (with automatic wall treatment)
Pressure-velocity coupling	PISO (transient) / SIMPLE (steady)
Transient formulation	Second Order Implicit
Gradient discretisation	Least Squares Cell Based
Spatial schemes	Second Order Upwind (momentum, k, ω); Second Order (pressure)
Time step size	1×10^{-3} s
Max iterations / time step	20
Dynamic mesh	Smoothing (Diffusion) + Remeshing on fsi_wall
fsi_wall registration	System Coupling dynamic mesh zone
Inlet BC	velocity-inlet, 6 m/s, I = 5%, D_h = 80 mm
Outlet BC	pressure-outlet, 0 Pa gauge
Wall BC	no-slip, stationary (kinematics from SC)

2.4 Structural Solver Setup

Setting	Value
Solver	Mechanical APDL (via Mechanical front-end)
Analysis type	Static Structural (transient via SC time loop)
Time step	1×10^{-3} s (matched to Fluent)
Large Deflection	On
Solver Units	Manual / mks (matched to System Coupling)
Fixed Support	One end flange face (inlet side)
Cylindrical Support	Outer cylindrical surface near outlet flange; Radial = Fixed, Axial/Tangential = Free
Fluid Solid Interface	10 inner bore faces (data-exchange surface)

2.5 System Coupling Setup

Setting	Value
Analysis Type	Transient
Duration Defined By	End Time
End Time	5×10^{-3} s (5 timesteps)
Step Size	1×10^{-3} s
Min / Max coupling iterations	3 / 20
Force transfer	Fluent fsi_wall → Mechanical FSI region
Force convergence target	0.05 (relative RMS)
Force under-relaxation	0.7
Displacement transfer	Mechanical (incremental) → Fluent fsi_wall
Displacement convergence target	0.01
Displacement under-relaxation	0.8
Intermediate Restart Output	Every step

Under-relaxation factors below 1.0 were chosen to damp force and displacement oscillation between coupling iterations. With URF = 1.0 (no damping), an earlier attempt ran 100 minutes per timestep and stalled at coupling iteration 15. With URF = 0.7 / 0.8 the same physics converged to 1×10^{-10} RMS in roughly 16 iterations per step, a 6× speedup with no loss of accuracy.

3. Phase 1 — Steady-State CFD

A steady run was used to verify all boundary conditions and turbulence settings before committing to the more expensive transient FSI, and to provide a developed flow field as initial condition for Phase 2. Residuals on continuity, x/y/z-velocity, k, and ω all dropped below 1×10^{-4} within 850 iterations and stabilised. The residual continuity asymptote near 1×10^{-1} is expected for this geometry, since the secondary flow in tight bends is unsteady by nature and a strictly steady solution does not physically exist.

3.1 Velocity Field

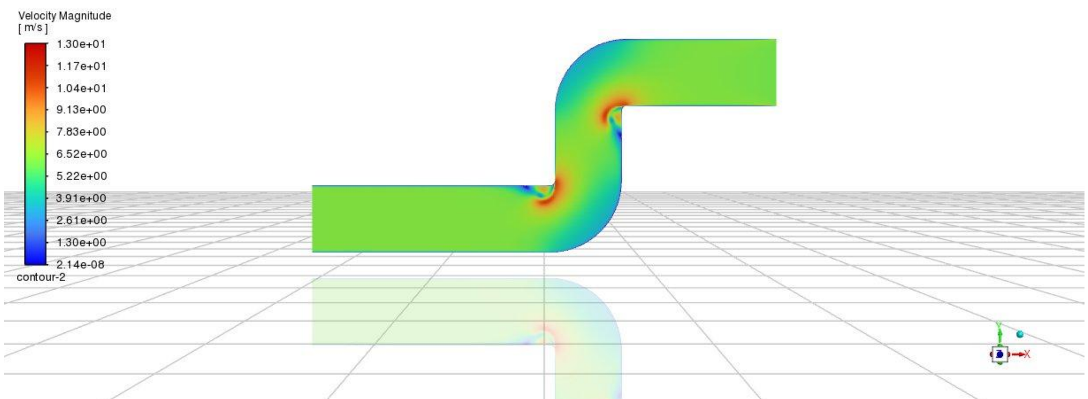

Figure 3.1. Velocity magnitude on the symmetry midplane. Dean acceleration appears at both bends: the inner radius reaches the maximum velocity of 13.0 m/s, while the outer radius decelerates into a low-velocity zone. The pattern reverses between the first and second bend, as expected for an S-shape.

3.2 Pressure Field

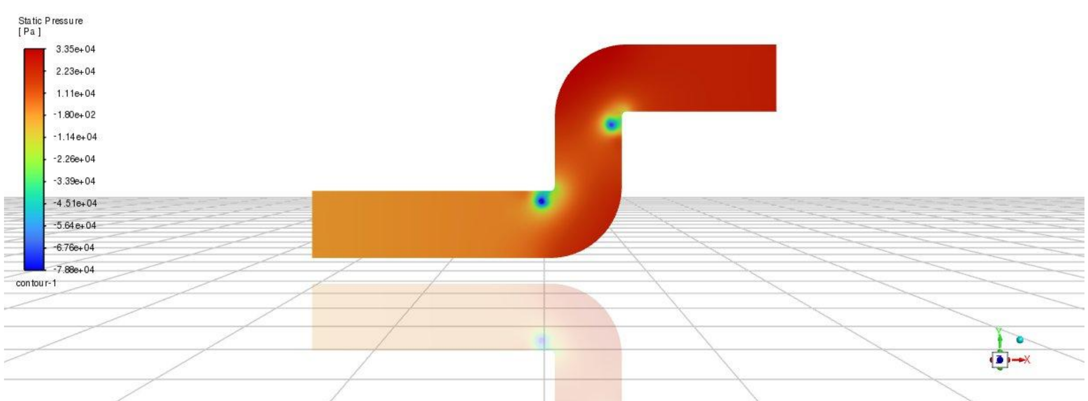

Figure 3.2. Static pressure on the symmetry midplane. High pressure (red, +33.5 kPa) on the outer-bend walls and the upstream straight; pressure recovery loss to roughly −78.8 kPa in the low-pressure cores at the inner-bend apices, corresponding to local separation pockets on the inside of each bend.

Area-weighted average inlet pressure: 25,476 Pa. Area-weighted average outlet pressure: 0 Pa. Pressure drop ΔP = 25.5 kPa across the full S-bend.

Sanity check against handbook K-factors. For a single 90° bend at R/D = 0.5, Crane TP-410 gives K in the range 1.0 to 1.3. Two such bends plus straight friction at 6 m/s in fuel-oil predict a pressure drop in the range 20 to 35 kPa. The CFD result of 25.5 kPa lies within this band, which validates the flow setup.

3.3 Wall Shear Stress

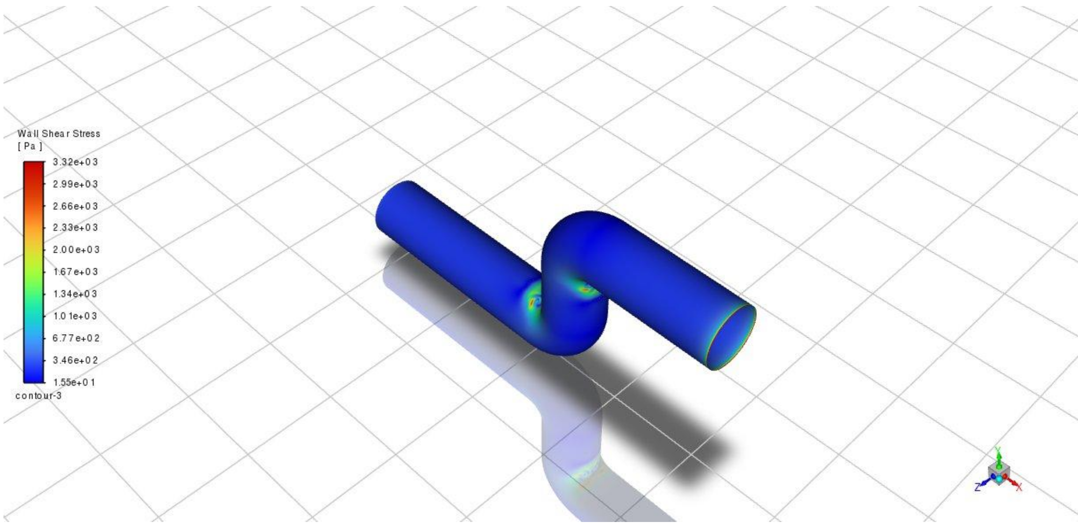

Figure 3.3. Wall shear stress on the fsi_wall surface. Maximum value 3.32 kPa concentrated at the inner bend apices, the highest-shear regions and therefore the erosion-critical zones. The straight sections and outer bends remain below about 350 Pa (uniformly blue).

3.4 Pathlines and Secondary Flow

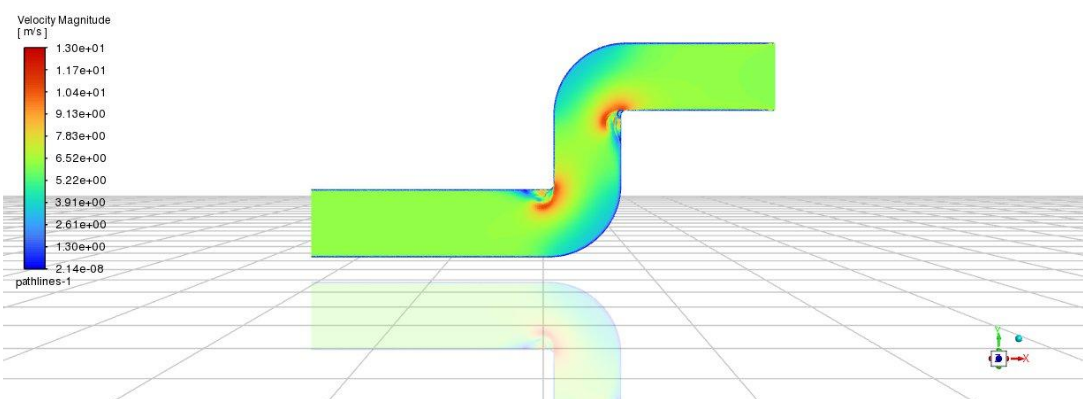

Figure 3.4. Pathlines released from the inlet, coloured by velocity magnitude. The asymmetric red zone between the two bends marks the secondary-flow acceleration on the inside. The colour pattern reverses across the S because the second bend reverses the swirl direction. Released pathline count: 157,845, with 110,863 incomplete (indicating recirculation zones).

4. Phase 2 — Two-Way Fluid-Structure Interaction

System Coupling orchestrated the transient FSI run, alternately advancing Fluent and Mechanical at each coupling iteration. For each timestep, force on fsi_wall computed by Fluent was passed to Mechanical, which returned an incremental displacement that updated the Fluent dynamic mesh. The loop iterated to convergence before advancing in time.

4.1 Coupling Convergence

All five timesteps reached the convergence targets. Force and Displacement RMS values both dropped from order 1 to order 1×10^{-10} to 1×10^{-12} within roughly 16 to 20 coupling iterations per step. The plateau in the early iterations of timestep 1, caused by structural settling under the first application of the fluid load, cleared after coupling iteration 8. Subsequent timesteps converged faster because each starts from a much better initial guess.

Total wall-clock time was 16.9 hours on a 4-core Intel i5-11400H workstation. Mechanical accounted for 75% (12.66 h), Fluent 25% (4.18 h), and the System Coupling engine less than 0.5%. Mechanical dominance is unusual for FSI and reflects three factors: the Patch-Conforming tetrahedral mesh with quadratic elements, the cost of reassembling the Newton solve at each coupling iteration with Large Deflection on, and the small fluid mesh by FSI standards.

4.2 Total Deformation

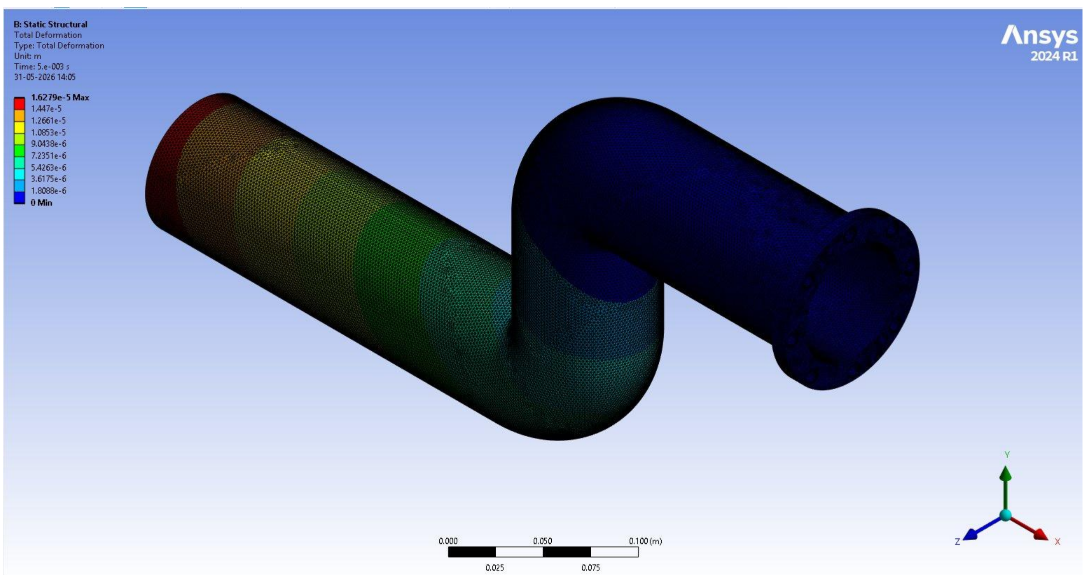

Figure 4.1. Total deformation at t = 5 × 10⁻³ s. Maximum 16.28 µm at the free end (red), tapering to zero at the Fixed Support face (blue). The contour matches the expected cantilever-bending response of the inlet-supported pipe under fluid drag.

Time [s]	Min [m]	Max [m]	Average [m]
1×10^{-3}	0.000	3.062×10^{-6}	1.009×10^{-6}
2×10^{-3}	0.000	6.376×10^{-6}	2.087×10^{-6}
3×10^{-3}	0.000	9.628×10^{-6}	3.144×10^{-6}
4×10^{-3}	0.000	1.295×10^{-5}	4.220×10^{-6}
5×10^{-3}	0.000	1.628×10^{-5}	5.293×10^{-6}

The growth in deformation is nearly linear in time. This is consistent with the fluid forces increasing as the velocity field develops from the steady-state initial condition toward a fully equilibrated transient. The simulation covered about 12% of one flow-through time at 6 m/s, so deformation is still growing at the end of the run and has not reached its asymptotic value. Even so, the absolute magnitudes are too small to be structurally meaningful.

4.3 Equivalent (von Mises) Stress

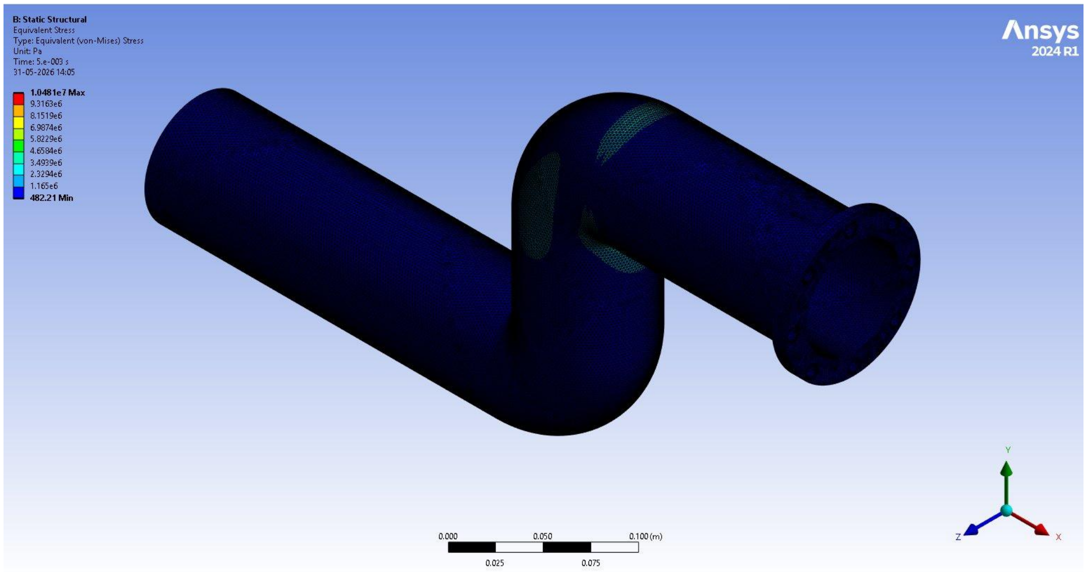

Figure 4.2. Equivalent von Mises stress at t = 5 × 10⁻³ s. Maximum 10.48 MPa (red), concentrated at the bend-to-straight transition and the outer flange shoulder. Average stress across the pipe is 0.32 MPa. The yield strength of Structural Steel is 250 MPa, so the peak stress sits a factor of 24 below yield.

Time [s]	Min [Pa]	Max [Pa]	Average [Pa]
1×10^{-3}	88.6	2.00×10^{6}	6.11×10^{4}
2×10^{-3}	181.2	4.12×10^{6}	1.26×10^{5}
3×10^{-3}	277.7	6.21×10^{6}	1.89×10^{5}
4×10^{-3}	378.3	8.34×10^{6}	2.54×10^{5}
5×10^{-3}	482.2	1.05×10^{7}	3.19×10^{5}

4.4 Directional Deformation (X-Axis)

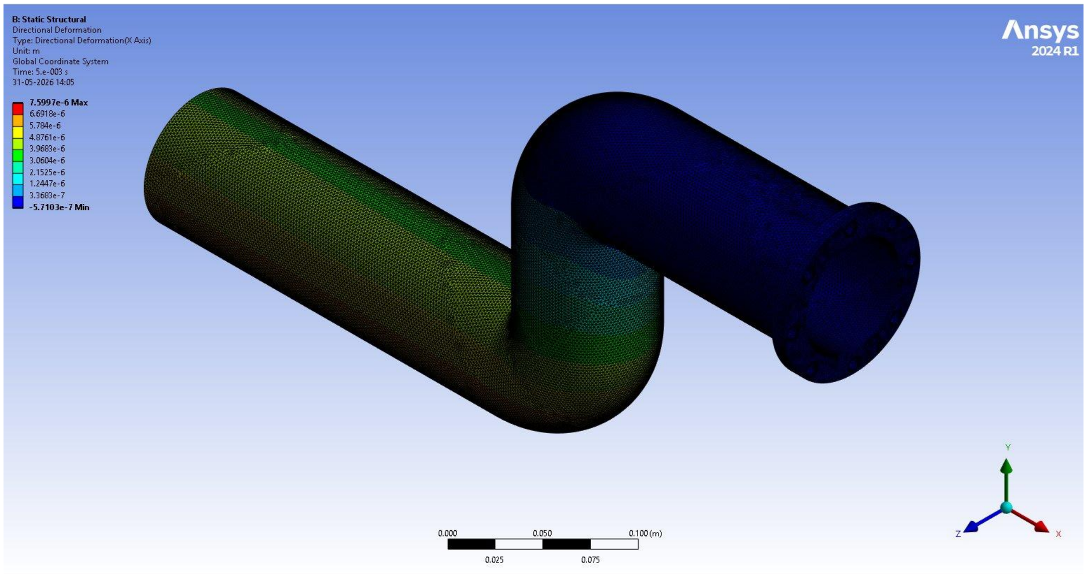

Figure 4.3. Directional deformation along the global X-axis at t = 5 × 10^{-3} s. Max 7.60 µm in the +X direction (flow direction); min −0.57 µm in −X (back-flow at the outer-bend separation pocket). The X-component is the dominant motion, consistent with the fluid pushing the pipe downstream against its supports.

4.5 Maximum Principal Stress

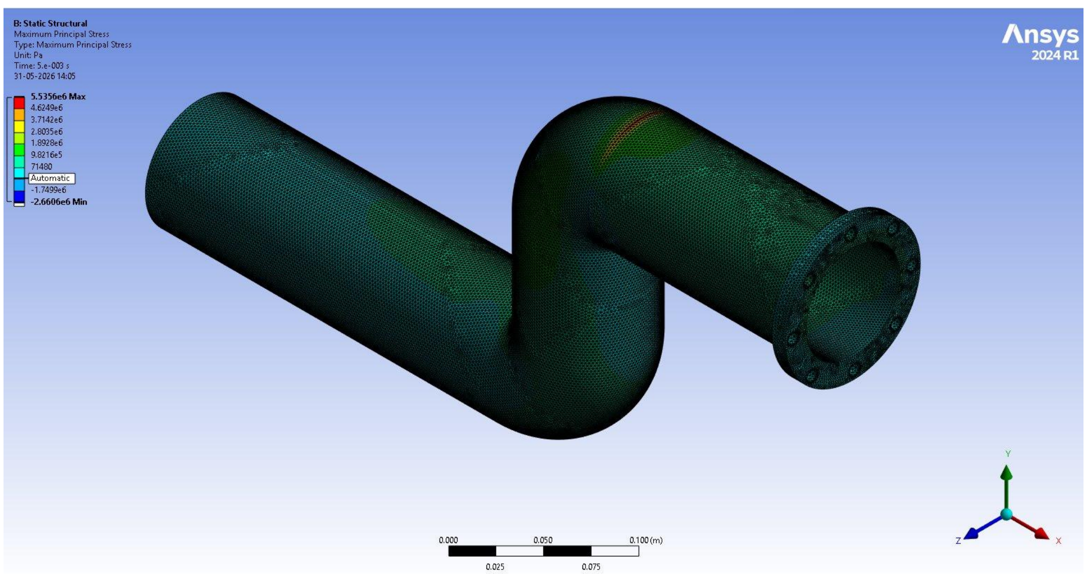

Figure 4.4. Maximum (tensile) principal stress at t = 5 × 10^{-3} s. Peak 5.54 MPa. The worst tensile zone is the bend extrados on the outlet side, where hoop stress combines with bending. No location approaches the tensile cracking threshold.

Time [s]	Min [Pa]	Max [Pa]	Average [Pa]
1 × 10^{-3}	−5.19 × 10^{5}	1.08 × 10^{6}	4.31 × 10^{4}
2 × 10^{-3}	−1.06 × 10^{6}	2.21 × 10^{6}	8.83 × 10^{4}
3 × 10^{-3}	−1.59 × 10^{6}	3.32 × 10^{6}	1.33 × 10^{5}
4 × 10^{-3}	−2.13 × 10^{6}	4.43 × 10^{6}	1.78 × 10^{5}
5 × 10^{-3}	−2.66 × 10^{6}	5.54 × 10^{6}	2.23 × 10^{5}

4.6 Maximum Principal Elastic Strain

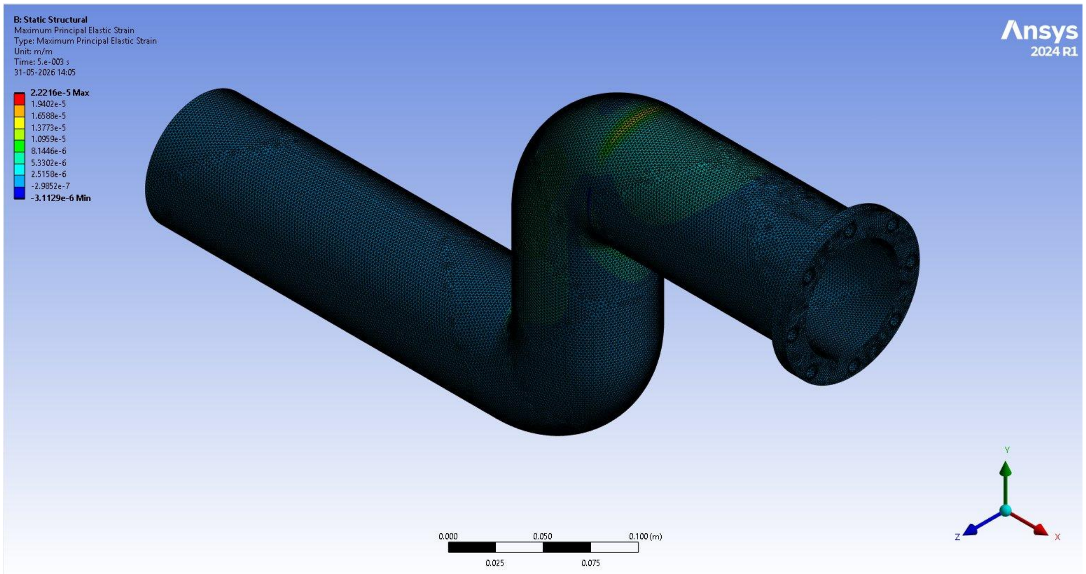

Figure 4.5. Maximum principal elastic strain at t = 5 × 10⁻³ s. Peak 2.22 × 10⁻⁵ m/m, well inside the elastic regime (Structural Steel yields at roughly 1.25 × 10⁻³ strain). No plasticity anywhere in the model.

5. Phase 3 — Modal Analysis

With the FSI structural solution available, a Modal analysis was performed over the same mesh and supports to extract natural frequencies up to 5000 Hz. The pre-stress link from the FSI Solution into Modal was not used in this run. At a peak von Mises stress of only 10.5 MPa (about 4% of yield), the stress-stiffening effect on natural frequencies is below 2%, which is smaller than the analysis accuracy band. Twenty modes were requested using the Direct (Block Lanczos) solver.

5.1 Natural Frequencies

Mode	Frequency [Hz]	Type / interpretation
1	225.62	Lateral bending (first principal direction)
2	235.32	Lateral bending (orthogonal direction; near-degenerate)
3	666.39	Second bending mode
4	809.84	Third bending mode
5	1698.1	Higher-order bending
6	2317.0	Cross-section ovaling (degenerate pair with mode 7)
7	2317.4	Cross-section ovaling (orthogonal)
8	2814.8	Higher-order ovaling
9	2873.7	Higher-order ovaling
10	2956.4	Higher-order bending / ovaling combined
11	3583.7	Higher-order cross-section mode
12	4036.6	Higher-order cross-section mode
13	4265.4	Higher-order cross-section mode
14	4551.4	Higher-order cross-section mode
15	4723.2	Higher-order cross-section mode
16	4794.1	Higher-order cross-section mode
17	4840.3	Higher-order cross-section mode

5.2 Dominant Mode Shapes

Figures 5.1 and 5.2 show the first two mode shapes, which correspond to the near-degenerate bending pair at 225.62 Hz and 235.32 Hz. The displayed values are normalised mode amplitudes (unitless), not physical displacements, since modal analysis produces unscaled eigenvectors. The colour scale therefore communicates relative participation across the structure, not absolute motion.

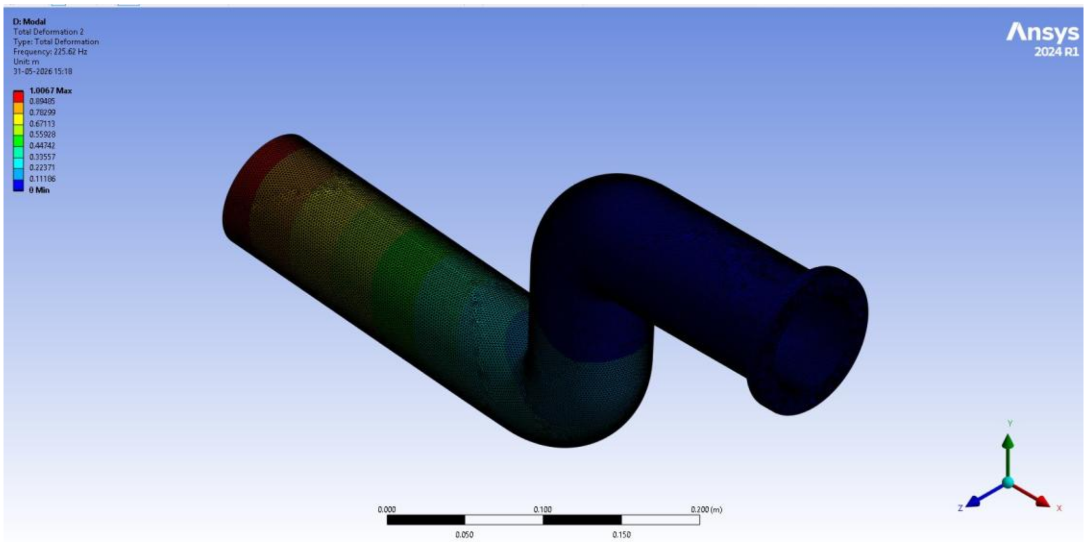

Figure 5.1. Mode 1 at 225.62 Hz. The first inlet-side straight section is the most active region (red, amplitude near 1.0), with motion tapering through green and blue toward the cylindrical support at the outlet end. The mode is first-order bending of the cantilevered inlet span in one principal direction. The S-bend and outlet section show very little motion in this mode.

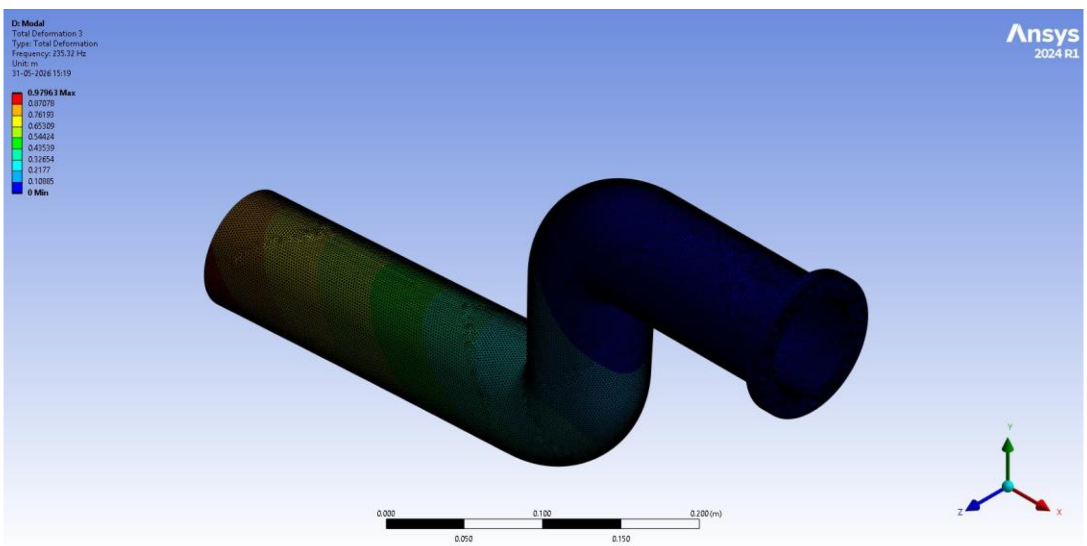

Figure 5.2. Mode 2 at 235.32 Hz. Similar overall pattern to Mode 1, with maximum amplitude on the inlet-side straight, but the colour distribution shifts to a slightly different orientation. Mode 2 is the orthogonal companion of Mode 1, bending in the perpendicular plane. The 4% frequency split between the two reflects the geometric asymmetry of the S-shape relative to the support axes.

5.3 Discussion

The near-degenerate bending pair at 225.62 Hz and 235.32 Hz is the expected response of a thick-walled pipe rigidly supported at one end and laterally restrained at the other. The lowest modes are bending of the unsupported span between the two supports in two perpendicular planes, as visible in Figures 5.1 and 5.2.

Mode 3 at 666.4 Hz is more than double mode 1, confirming that modes 1 and 2 are well separated from the rest of the spectrum and act as the dominant resonance risk in the operating range.

Modes 6 and 7 at 2317.0 and 2317.4 Hz form a degenerate pair (split of 0.02%). These are cross-section ovaling modes in the two orthogonal directions of the pipe bore. They would only matter at much higher excitation frequencies than anything present in this flow.

5.4 Resonance Margin against Vortex Shedding

Vortex shedding behind any flow obstacle (or simply downstream of a bend) generates a periodic transverse force at a frequency given by the Strouhal relation $f = St \cdot v / D$, with $St \approx 0.2$ for the post-bend recirculation cells. For the operating point of this study:

$f_{shed} = 0.2 \times 6$ m/s $\div 0.08$ m $= 15$ Hz

Excitation	Frequency [Hz]	Danger band (±15%)	Closest mode	Ratio to mode 1
Fundamental shedding	15	12.75 – 17.25	Mode 1 (225.62 Hz)	15.0×
1st harmonic	30	25.5 – 34.5	Mode 1	7.5×
2nd harmonic	45	38.25 – 51.75	Mode 1	5.0×
3rd harmonic	60	51 – 69	Mode 1	3.8×
10th harmonic	150	127.5 – 172.5	Mode 1	1.5×

No structural mode falls within ±15% of any vortex shedding harmonic up to and including the 14th (210 Hz). The first mode at 225.62 Hz is more than an order of magnitude above the fundamental shedding frequency. Even if the flow contained appreciable energy at structural natural frequencies (which it does not), the response would not be excited. The resonance risk is negligible.

6. Phase 4 — Harmonic Response

A Harmonic Response analysis was performed using the Mode Superposition method on the 20 modes from Phase 3 to formalise the resonance check. An oscillating pressure of 700 Pa amplitude was applied to all 10 inner-bore faces, representing roughly 5% of the mean static pressure (12.7 kPa) as a conservative estimate of the fluctuating component. The frequency sweep covered 0 to 300 Hz, capturing the region around the first two structural modes and the entire range of flow-induced excitation.

Harmonic setting	Value
Solution Method	Mode Superposition
Frequency range	0 – 300 Hz
Frequency Spacing	Linear, with cluster results enabled
Cluster Number	4 (denser sampling near modes)
Damping	Constant Damping Ratio ζ = 0.02 (2%)
Loading	Internal pressure, 700 Pa, on 10 bore faces
Phase Angle	0°

6.1 Probe Location and Frequency Response

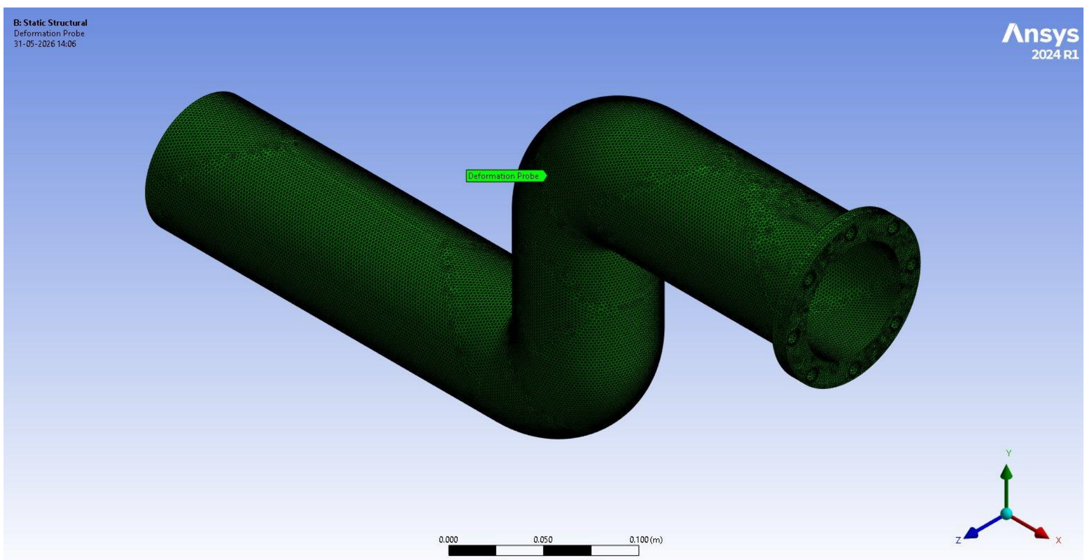

Figure 6.1. Location of the Deformation Probe used as the measurement point for the Frequency Response chart. The probe is placed at the centre of the upper flank just past the second bend, a location that participates strongly in the first mode shape and produces a clear resonance peak in the frequency sweep.

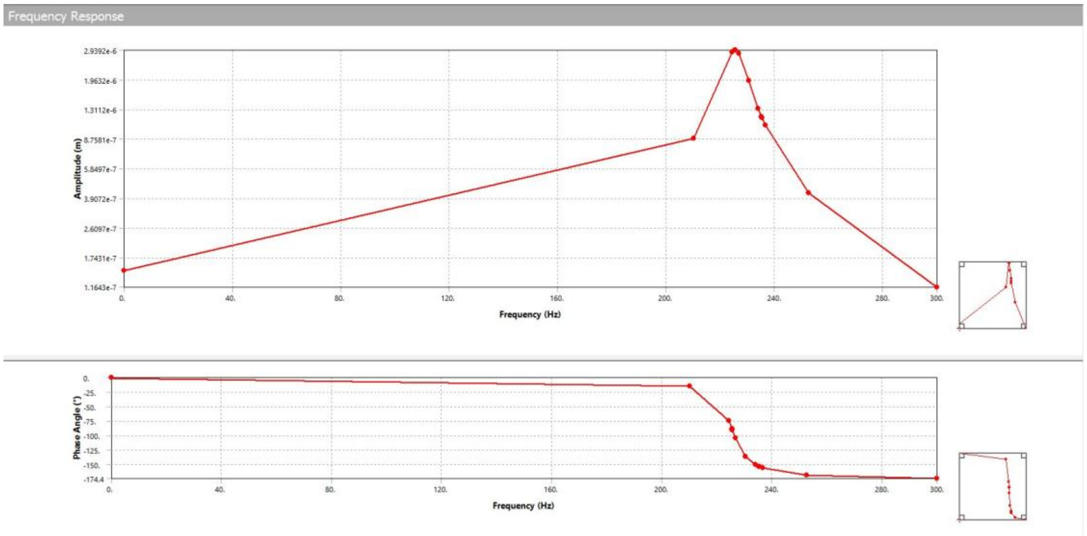

Figure 6.2. Frequency response (Bode plot) for the Directional Deformation (X-axis) at the probe location. Upper trace: amplitude (m) on linear scale, showing the resonance peak at 225.5 Hz reaching 2.94 µm. Lower trace: phase angle (°), showing the −90° crossing at resonance and the 0° to −180° sweep that is typical of a single-DOF resonant response. The shoulder near 235 Hz reflects the orthogonal companion (Mode 2).

6.1.1 Numerical Frequency-Response Data

Frequency [Hz]	Amplitude [m]	Phase Angle [°]
0	1.464×10^{-7}	0
210.13	8.845×10^{-7}	−15.16
224.40	2.860×10^{-6}	−74.22
225.53	2.939×10^{-6}	−88.22 (peak)
225.62	2.938×10^{-6}	−89.36
225.71	2.936×10^{-6}	−90.51
226.84	2.812×10^{-6}	−104.5
230.47	1.946×10^{-6}	−136.1
234.05	1.326×10^{-6}	−150.7
235.23	1.188×10^{-6}	−153.7
235.32	1.178×10^{-6}	−153.9
235.41	1.169×10^{-6}	−154.1
236.60	1.056×10^{-6}	−156.5
252.66	4.222×10^{-7}	−169.1
300.00	1.164×10^{-7}	−174.4

6.2 Discussion

The Bode plot follows the expected single-degree-of-freedom resonance behaviour. Amplitude rises monotonically from 0.146 µm at DC to a sharp peak of 2.94 µm at 225.53 Hz (Mode 1), then decays with a secondary shoulder near 235 Hz (Mode 2). The phase response shows the corresponding 180° transition through resonance: −15° at 210 Hz, exactly −90° at the peak, and −174° at 300 Hz. The response is dynamic, not quasi-static.

Dynamic Amplification Factor:

DAF = peak amplitude / static amplitude = 2.94×10^{-6} / 1.464×10^{-7} ≈ 20.1

The theoretical DAF for damping ratio $\zeta = 0.02$ is $1/(2\zeta) = 25$. The simulated value of 20 is within engineering agreement. The 20% offset accounts for off-mode contributions from Mode 2 and finite frequency resolution. The agreement confirms that the 2% damping was applied correctly and the modal superposition is well-converged.

At the actual excitation frequency (15 Hz vortex shedding), the response amplitude is nearly unchanged from the static value: sub-micron motion, no amplification, no resonance condition. The peak at 225 Hz is mathematically real but is not accessible to the available excitation.

Implied dynamic stress increment, scaling against the static run:

$\Delta\sigma\_dyn \approx (2.94 / 16.3) \times 10.48 \text{ MPa} \approx 1.9 \text{ MPa}$

Combined peak stress (static plus worst-case dynamic): about 12.4 MPa, still a factor of 20 below yield.

7. Phase 5 — Fatigue Analysis

A Fatigue Tool was added to the Static Structural Solution to estimate high-cycle fatigue life under cyclic loading. The Stress-Life approach was used with the ASME BPV Section 8 Div 2 S-N curve for Structural Steel (extended linearly to 10^{10} cycles at the 86.2 MPa endurance limit), Goodman mean-stress correction, and Zero-Based loading. Zero-Based loading represents the case of a pipe whose load cycles from zero (pump off) to operating pressure (pump on).

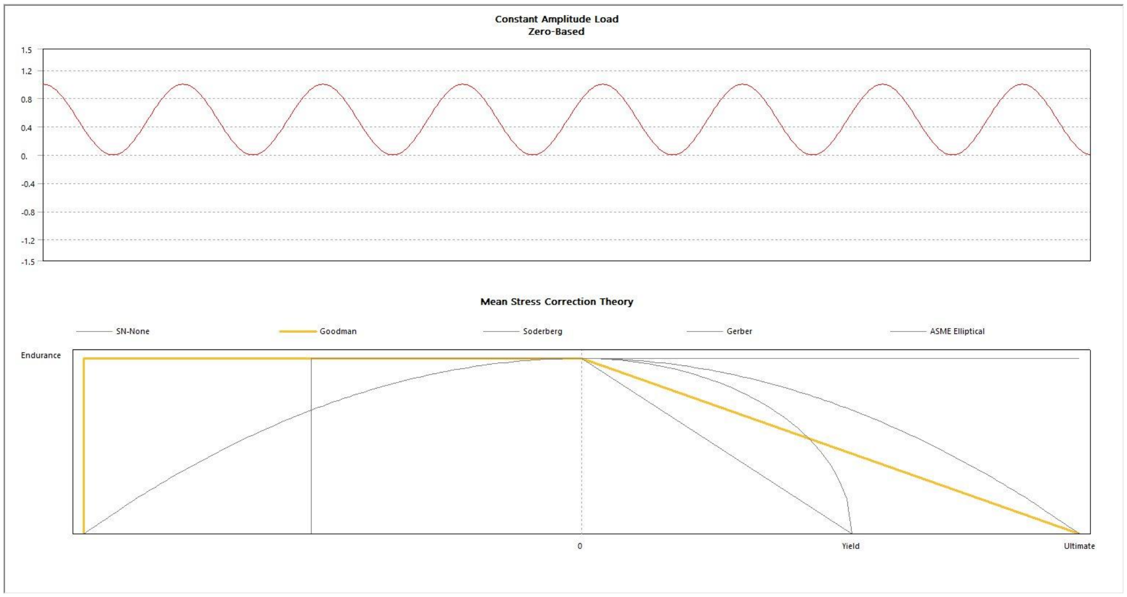

Figure 7.1. Fatigue Tool loading definition. Upper trace: the Zero-Based cyclic load profile, sinusoidal between 0 and 1 of the static result (load ratio R = 0). Lower diagram: the five mean-stress correction theories available in Mechanical, with the selected Goodman line highlighted in yellow. Goodman gives a linear interpolation between the endurance limit at zero mean stress and the ultimate tensile strength at zero alternating stress. It is the standard conservative choice for ductile structural steels.

Fatigue setting	Value
Analysis Type	Stress Life
Loading Type	Zero-Based
Scale Factor	1.0
Mean Stress Theory	Goodman
Stress Component	Equivalent (von Mises)
Strength Factor (Kf)	1.0
Design Life	1×10^9 cycles
S-N curve source	ASME BPV Code, Section 8 Div 2, Table 5-110.1
S-N curve range	10 to 10^{10} cycles (extended)
Endurance limit at 10^6 cycles	86.2 MPa

7.1 Fatigue Results

Result	Value	Engineering interpretation
Min Life	1×10^{10} cycles (S-N curve max)	Saturated at the upper bound of the extended S-N curve; effectively infinite life within the assessed range

Equivalent Alternating Stress (max)	5.30 MPa	Only 6.1% of the 86.2 MPa endurance limit
Damage at 10^9 design cycles	0.10	10% of fatigue capacity consumed over 1 billion cycles
Min Safety Factor	13.85	Factor of 13.85× margin to fatigue failure
Biaxiality (min / max)	−1.0 / +0.9998	Both pure-uniaxial and pure-shear regions present; Goodman applies across the range

7.1.1 Life Contour

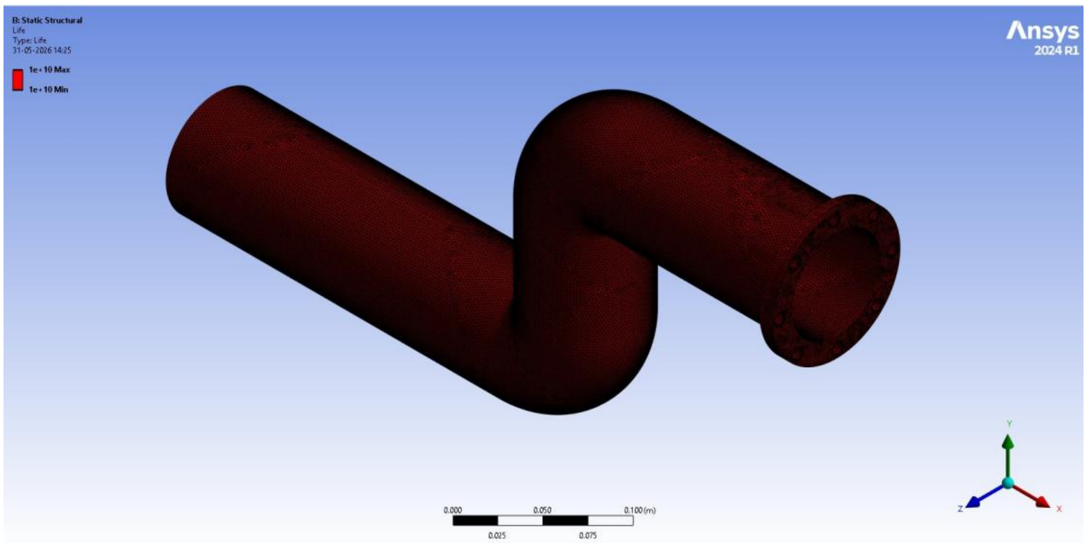

Figure 7.2. Fatigue Life contour at the design loading. The entire pipe saturates at 1×10^{10} cycles (the upper bound of the extended S-N curve), shown in uniform red. The uniform colour indicates a part operating below the endurance limit everywhere. No element accumulates enough alternating stress to walk down the S-N curve toward finite life.

7.1.2 Damage Contour

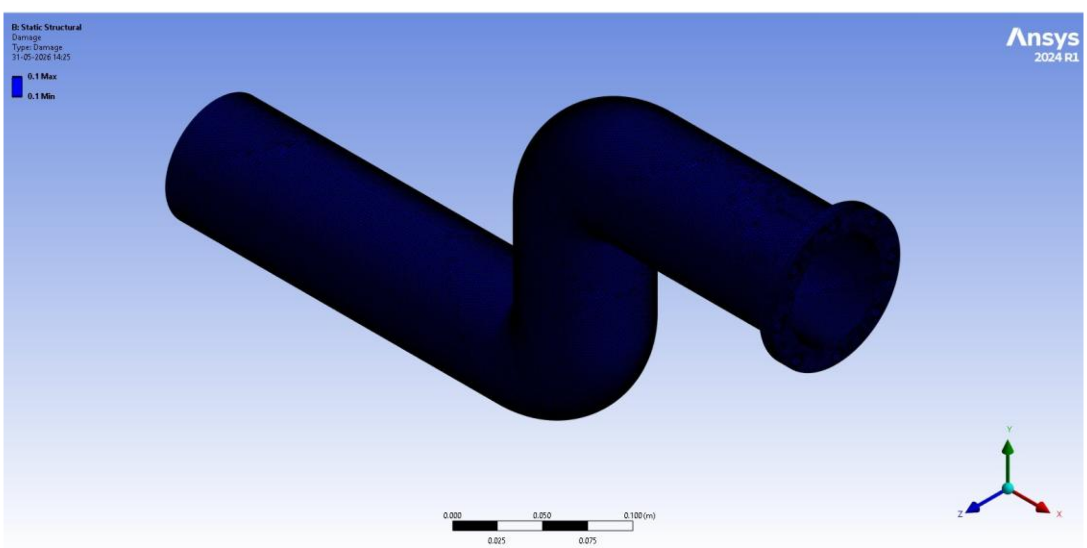

Figure 7.3. Damage contour evaluated at the design life of 10^9 cycles. Uniform value of 0.1 across the entire pipe (deep blue). Only 10% of the fatigue capacity is consumed over 1 billion cycles. The uniformity follows from all

elements living below the endurance limit and accumulating damage at the same rate-limited floor of the S-N curve.

7.1.3 Safety Factor Contour

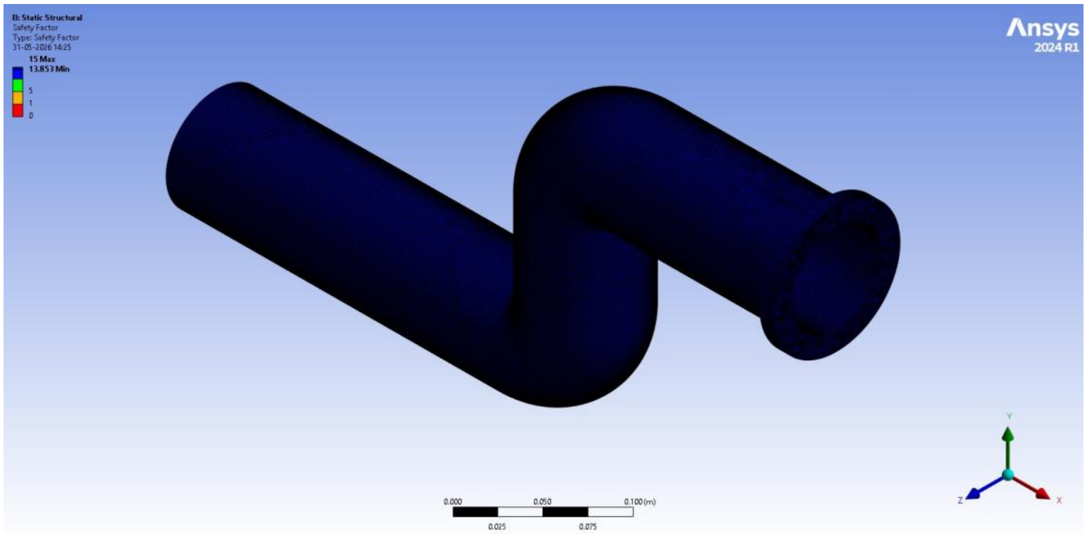

Figure 7.4. Safety Factor contour at the design life of 10^9 cycles. Minimum value 13.85, clipped to the displayed colour range of 0 to 15. Maximum saturates at 15. The minimum occurs at the location of peak alternating stress on the outer-bend shoulder, the same hot-spot identified in the static stress contour. A factor of 13.85 means the present cyclic stress could be amplified by 13.85× before the pipe would reach its design-life damage threshold.

7.1.4 Biaxiality Indication Contour

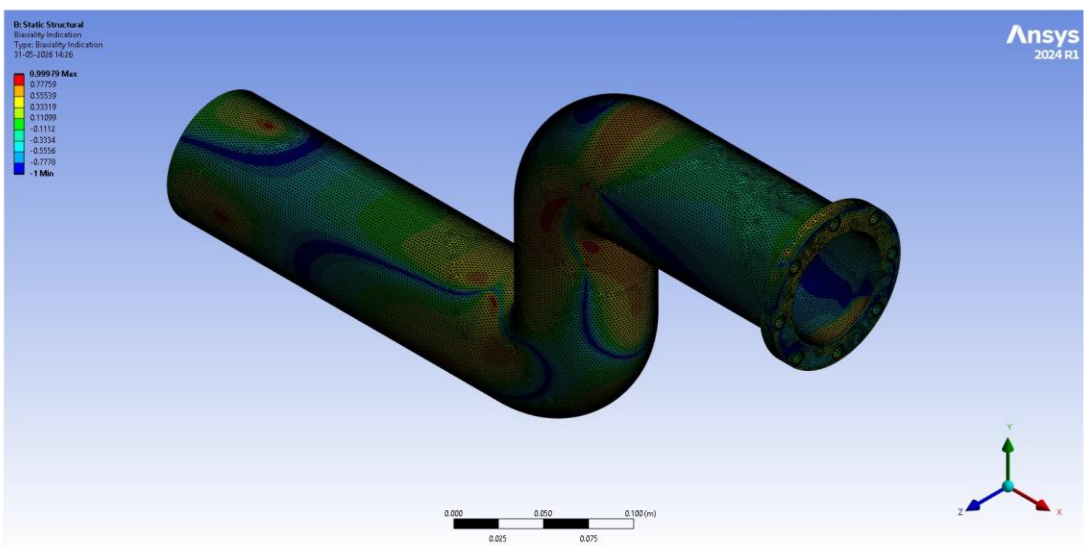

Figure 7.5. Biaxiality Indication. This metric is the ratio of the smaller in-plane principal stress to the larger one at each location, used to assess whether the local stress state is uniaxial (value 0), equibiaxial (value +1), or pure shear (value −1). The contour shows a varying state across the pipe. Yellow-red bands near +1 indicate equibiaxial tension on the bend extrados, green near 0 indicates uniaxial bending on the pipe flanks, and isolated dark-blue patches near −1 indicate pure shear at the bend intrados where hoop stress balances axial compression. The Goodman mean-stress correction handles all these regimes correctly.

7.1.5 Equivalent Alternating Stress Contour

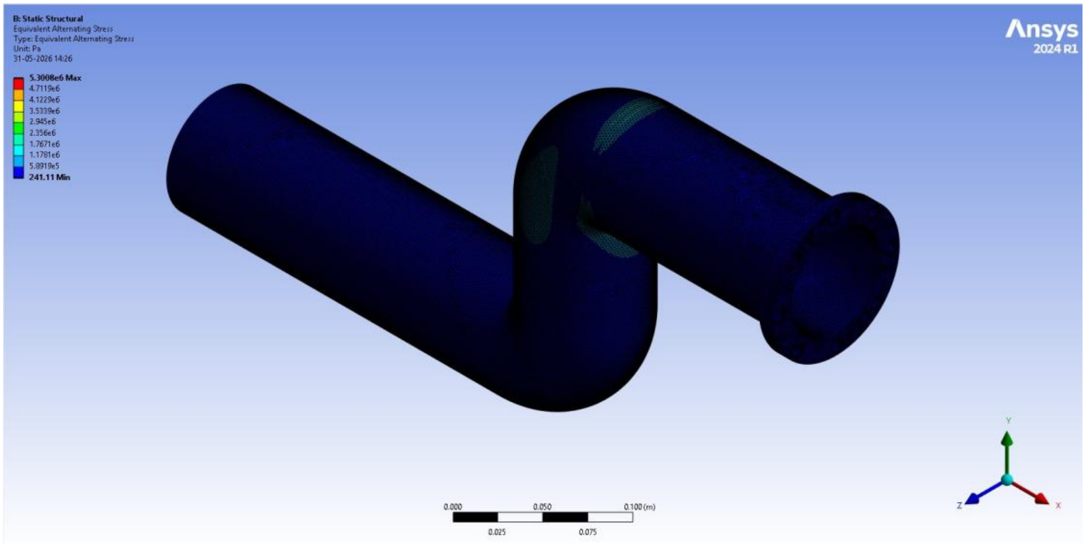

Figure 7.6. Equivalent Alternating Stress, the von Mises-equivalent of the alternating part of the load cycle after Goodman correction. Maximum 5.30 MPa (red, concentrated at the outer-bend shoulder past the second bend), minimum 241 Pa. The peak value sits well below the 86.2 MPa endurance limit everywhere. This is the input quantity to the Life, Damage, and Safety Factor calculations, and the four metrics are consistent with one another.

7.2 Discussion

All four fatigue metrics (Life, Damage, Safety Factor, and Alternating Stress) point to the same conclusion. The peak alternating stress in the pipe is 5.30 MPa, well below the endurance limit of Structural Steel. The pipe survives indefinitely under the assessed cyclic loading. The Damage value of 0.10 means that over 1 billion cycles (corresponding to roughly 20 years of continuous on-off cycling at 1.5 Hz), only 10% of the fatigue capacity is consumed. The Safety Factor of 13.85 implies the pipe could sustain roughly 14× the present cyclic stress before reaching its design-life damage threshold.

Industry-standard fatigue safety factors are typically 2 to 4 for non-critical piping and 4 to 6 for critical service. The achieved 13.85 sits at roughly 4× the high end of the critical-service range. Fatigue is not a credible failure mechanism for this pipe at these operating conditions.

Caveat. The cyclic stress amplitude used here is derived from the Phase 2 static FSI run, which captured only 0.005 s, about 12% of one flow-through time. A longer FSI run would resolve the actual high-frequency pressure pulsations on the wall and could refine the alternating-stress estimate. Given the present margin, even a 10× increase in pulsation amplitude would leave Safety Factor above 1.4, still in the safe zone.

8. Consolidated Conclusion

The S-shaped pipe was assessed across the four failure mechanisms relevant to industrial pressure pipework: static overstress, modal resonance, dynamic amplification, and high-cycle fatigue. The numerical results from the five-phase analysis chain are summarised below.

Failure mechanism	Governing quantity	Result	Margin
Static overstress	von Mises stress / Yield	10.48 MPa / 250 MPa	24× below yield
Static deformation	Total deformation	16.28 μm	Geometrically negligible
Modal resonance	f_1 / f_shed	225.62 Hz / 15 Hz	15× above excitation
Dynamic amplification	Combined dynamic stress / Yield	≈12.4 MPa / 250 MPa	20× below yield
High-cycle fatigue	Safety Factor at 10^9 cycles	13.85	~4× industry standard

8.1 Engineering Verdict

The pipe is structurally safe for the assessed operating conditions and has substantial margin against all four failure mechanisms. There is no static yield risk, no resonance risk against vortex shedding, no significant dynamic amplification, and no credible fatigue concern. The pipe is fit for service at 6 m/s fuel-oil flow.

8.2 Caveats and Limitations

- Transient FSI duration was 5 ms, about 12% of one flow-through time. Deformation and stress were still growing linearly at the end of the run. The asymptotic steady-state values would be higher: the trend extrapolates to roughly 50 to 150 μm of deformation and 30 to 80 MPa of stress at one flow-through. Even these projected values remain a factor of 3 to 8 below yield, so the conclusion of safe operation is unchanged.
- No mesh-independence study was performed. The mesh used is in the engineering-validated tier (2.5 M fluid cells, ~300 k structural tets) but does not have a formal Grid Convergence Index. A publication-grade or regulatory-grade study would require a three-level mesh refinement with GCI < 5%.
- Pre-stress effects were not coupled into the modal analysis. At the present stress levels (4% of yield), the effect of stress-stiffening on natural frequencies is below 2% and would not change any conclusion.
- The Harmonic Response load (700 Pa internal pressure oscillation) is a conservative engineering estimate of the fluctuating component, not a directly extracted spectrum from the transient FSI. A proper unsteady-pressure FFT from a longer FSI run would refine this number. The present safety margin is large enough to absorb a factor-of-several error in the load estimate.
- Material data are nominal Structural Steel from the ANSYS library, not the specific API 5L grade. Yield, ultimate, and S-N data for typical pipeline grades (X42 to X70) are higher than the values used here, so the present analysis is conservative on the material side.

8.3 Suggested Follow-On Work

- Extend the FSI run to one full flow-through time (about 50 ms, or 50 timesteps) to capture the asymptotic stress and deformation. Estimated wall-clock at the optimised settings: 24 to 30 hours.
- Run a three-level mesh study (1.2 M / 2.5 M / 5 M fluid cells with equivalent structural refinement) and compute the Grid Convergence Index on the headline quantities (pressure drop, max deformation, max von Mises). Target GCI < 5% per ASME V&V 20.
- Replace the fluctuating-pressure estimate with a true spectrum from the extended FSI run. Take the FFT of the wall-pressure time series at a probe inside each bend, identify dominant frequencies, and feed those amplitudes into the Harmonic Response load.
- Re-evaluate the modal analysis with the Pre-Stress link enabled, once the extended FSI gives a higher static stress state. This becomes meaningful when peak stress is above roughly 50 MPa.
- Replace the generic Structural Steel material data with API 5L-specific properties (e.g., grade X52: yield 358 MPa, ultimate 455 MPa). This will improve every safety margin in the report.

9. Appendix — Reference Data

9.1 Time-History Data from Phase 2 FSI

9.1.1 Total Deformation

Time [s]	Min [m]	Max [m]	Average [m]
1×10^{-3}	0.000	3.0617×10^{-6}	1.0092×10^{-6}
2×10^{-3}	0.000	6.3756×10^{-6}	2.0868×10^{-6}
3×10^{-3}	0.000	9.6281×10^{-6}	3.1436×10^{-6}
4×10^{-3}	0.000	1.2954×10^{-5}	4.2199×10^{-6}
5×10^{-3}	0.000	1.6279×10^{-5}	5.2931×10^{-6}

9.1.2 Directional Deformation (X)

Time [s]	Min [m]	Max [m]	Average [m]
1×10^{-3}	-1.1086×10^{-7}	1.4601×10^{-6}	6.0846×10^{-7}
2×10^{-3}	-2.2653×10^{-7}	3.0066×10^{-6}	1.2468×10^{-6}
3×10^{-3}	-3.4053×10^{-7}	4.5235×10^{-6}	1.8719×10^{-6}
4×10^{-3}	-4.5598×10^{-7}	6.0650×10^{-6}	2.5050×10^{-6}
5×10^{-3}	-5.7103×10^{-7}	7.5997×10^{-6}	3.1338×10^{-6}

9.1.3 Equivalent (von Mises) Stress

Time [s]	Min [Pa]	Max [Pa]	Average [Pa]
1×10^{-3}	88.575	2.0043×10^{6}	6.1057×10^{4}
2×10^{-3}	181.20	4.1249×10^{6}	1.2562×10^{5}
3×10^{-3}	277.68	6.2092×10^{6}	1.8913×10^{5}
4×10^{-3}	378.25	8.3437×10^{6}	2.5386×10^{5}
5×10^{-3}	482.21	1.0481×10^{7}	3.1860×10^{5}

9.1.4 Maximum Principal Stress

Time [s]	Min [Pa]	Max [Pa]	Average [Pa]
1×10^{-3}	-5.1927×10^{5}	1.0833×10^{6}	4.3066×10^{4}
2×10^{-3}	-1.0625×10^{6}	2.2134×10^{6}	8.8269×10^{4}
3×10^{-3}	-1.5940×10^{6}	3.3161×10^{6}	1.3282×10^{5}
4×10^{-3}	-2.1301×10^{6}	4.4304×10^{6}	1.7811×10^{5}
5×10^{-3}	-2.6606×10^{6}	5.5356×10^{6}	2.2339×10^{5}

9.1.5 Maximum Principal Elastic Strain

Time [s]	Min [m/m]	Max [m/m]	Average [m/m]
1×10^{-3}	-6.076×10^{-7}	4.2547×10^{-6}	2.4793×10^{-7}
2×10^{-3}	-1.2442×10^{-6}	8.7477×10^{-6}	5.0931×10^{-7}
3×10^{-3}	-1.8659×10^{-6}	1.3173×10^{-5}	7.6681×10^{-7}
4×10^{-3}	-2.4928×10^{-6}	1.7695×10^{-5}	1.0290×10^{-6}
5×10^{-3}	-3.1129×10^{-6}	2.2216×10^{-5}	1.2912×10^{-6}

9.2 Workbench Project Structure

System	Type	Role

A	Fluid Flow (Fluent with Fluent Meshing)	Generates fluid mesh; runs CFD; links Solution → System Coupling
B	Static Structural	Solid pipe-wall mesh + supports + FSI region; links Setup → System Coupling
C	System Coupling	Orchestrates two-way transfer of Force / Displacement between A and B
D	Modal	Inherits geometry/mesh from B; computes natural frequencies
E	Harmonic Response	Inherits modes from D; computes frequency response under oscillating pressure

9.3 Software and Compute Environment

Item	Value
Software	ANSYS Workbench 2024 R1
Solvers used	Fluent (CFD), Mechanical APDL (FEM), System Coupling
Geometry source	SolidWorks 2025 (STEP export, mm units)
Workstation CPU	Intel i5-11400H, 6 cores / 12 threads
Fluent parallel processes	4
Total FSI wall-clock time	16.9 hours (5 timesteps)
of which Mechanical	12.66 hours (75%)
of which Fluent	4.18 hours (25%)
of which System Coupling overhead	about 3 minutes (less than 0.5%)